Project Charter Document

Inventra – Inventory Management System

1. Purpose

The purpose of the Inventra system is to provide a **simple and efficient digital platform for small-to-medium building material retailers** to manage their daily business operations.

The system helps retailers to:

- Generate GST-compliant invoices quickly
- Track inventory levels in real time
- Maintain structured customer records
- Monitor sales performance
- Reduce manual calculation errors
- Improve operational efficiency

By replacing manual paper-based processes with a digital system, the platform improves business accuracy, productivity, and decision-making.

2. Objectives

The main objectives of the project are:

- Automate billing and GST calculation for retail businesses.
- Enable real-time inventory tracking and stock updates.
- Maintain centralized customer records and purchase history.
- Provide sales reports for better business insights.
- Reduce billing errors and manual workload.
- Digitize retail business operations to improve efficiency.

3. Demand / Opportunity

Many small and medium building material retailers still rely on **manual registers, handwritten bills, and spreadsheets** to manage their business.

This creates several problems:

- Incorrect GST calculations
- Difficulty tracking stock levels
- Misplaced customer records
- Slow billing process
- Lack of business insights

Existing ERP systems are often **too complex or expensive for small retailers**.

The Inventra system fills this gap by offering a **simple, affordable, and easy-to-use digital retail management platform** designed specifically for small retail businesses.

4. Project Scope Statement

4.1 In Scope (Version 1)

- Role-based authentication (Admin, Staff, Viewer)
- Product and category management
- Invoice generation with GST breakdown
- Real-time inventory updates
- Customer management
- Basic reporting (Daily / Monthly Sales, GST summary)
- Single-shop deployment model

4.2 Out of Scope (Future Versions)

- Multi-branch management

- Mobile application
- Supplier/vendor management
- Online payment integration
- Accounting software integration (e.g., Tally)
- Advanced analytics or AI forecasting
- Offline mode support

5. Assumptions

- The system will be deployed for a single retail shop instance.
- Users have basic computer literacy.
- Internet connectivity is available during business hours.
- GST rates remain stable during the development phase.
- Product catalog size will remain within small retail limits (under ~10,000 products).

6. Constraints

- Development timeline limited to 5 weeks.
- Limited development team size.
- Budget constraints restrict enterprise infrastructure.
- No dedicated DevOps or production support team.
- MVP release prioritizes functionality over advanced optimization.

7. Business Requirements

7.1 User Authentication

- Secure login using email and password

- Role-based access control (Admin, Staff, Viewer)

7.2 Billing Management

- Generate GST-compliant invoices
- Automatic GST calculation
- Print or download invoice
- Maintain invoice history

7.3 Inventory Management

- Add, edit, and delete products
- Track product stock levels
- Automatic stock updates after each sale
- Low stock alerts

7.4 Customer Management

- Add and manage customer records
- Store GSTIN details
- Maintain purchase history

7.5 Reporting and Analytics

- Daily sales report
- Monthly sales report
- GST summary report
- Stock reports

8. Technical Requirements

8.1 Authentication System

- Secure login and session handling using JWT tokens
- Password encryption using secure hashing algorithms

8.2 Billing & Inventory Processing

- Backend logic for invoice creation
- GST calculation module
- Inventory deduction after sales transactions

8.3 Reporting System

- Generate reports for sales and inventory
- Export reports in PDF or Excel format

8.4 Role-Based Access Control

- Admin: Full access
- Staff: Billing and customer management
- Viewer: Read-only access to reports

9. Technological Requirements

9.1 Frontend

- React.js
- TypeScript
- Tailwind CSS

9.2 Backend

- Spring Boot
- Spring Security
- REST APIs

9.3 Database

- PostgreSQL

9.4 Authentication

- JWT Token Authentication
- BCrypt password encryption

9.5 Development Tools

- Visual Studio Code
- Git & GitHub
- Postman

10. Stakeholders

Stakeholder	Role
Retail Shop Owners	Primary users of the system
Billing Staff	Generate invoices and manage customers
System Developers	Develop and maintain the system
Testers	Validate system functionality
Project Guide / Faculty	Provide project supervision

11. Resources Needed

11.1 Documentation

- React.js Documentation
- Spring Boot Documentation
- PostgreSQL Documentation
- REST API Development Guidelines

11.2 Human Resources

Role	Count	Responsibility
Product Owner	1	Defines requirements & priorities
Scrum Master	1	Facilitates sprints & removes blockers
Backend Developer	1-2	API, database, business logic
Frontend Developer	1-2	UI development & integration
QA Tester	1	Functional & integration testing

12. Risk Analysis

Risk	Description	Mitigation Strategy

Data Breach	Unauthorized access to sensitive data	Implement JWT authentication and encrypted passwords
System Downtime	Server or database failures	Use reliable cloud hosting
Internet downtime at shop	Internet lost due to some reasons	Manual fallback (future offline feature)
Security Vulnerability	Unauthorized system access	Role-based access control
User Interface Issues	Users may find the system complex	Conduct usability testing

13. Timeline / Milestones

Phase	Milestone Tasks	Timeline
Planning	Requirement analysis ,planning and system design	Week 1
Backend Development	Database design and API development	Week 2
Billing & Inventory Module	Invoice generation, GST calculation and main billing logic implementation	Week 3
Frontend Development & Integration	UI development, billing interface and API integration	Week 4
Testing & Deployment	Functional testing, bug fixing, deployment system finalization and documentation freeze	Week 5

14. Budget

Service	Estimated Cost	Type	Description
Cloud Hosting	₹5,000	Annual	Used for backend server deployment and application hosting.
Database Hosting	₹7,000	Annual	Managed PostgreSQL service for data storage.
Development Cost	₹100,000	One-Time	Used for backend server deployment and application hosting.
Miscellaneous	₹10,500	One-Time	Domain + Tools + Documentation
Total Estimated Cost	**₹1,21,500**		

15. RACI Chart

Task / Activity	Aditya Ubale	Vaibhav Kawa	Sahas Nagar	Shreya Newaskar	Ashwini Arude	Mrugmai Dudhamande
Requirement Analysis	R	C	I	C	A	R
System Architecture Design	I	C	R	I	A	R
Database Design (Products, Customers, Invoices)	C	A	I	R	R	A
Authentication & Security (JWT, RBAC)	R	A	R	C	C	I
Billing & GST Calculation Logic	R	R	A	I	I	C
Inventory Management Module	A	C	I	C	R	R
Customer Management Module	I	A	C	A	R	C
Frontend UI Development	I	A	C	R	A	C
API Integration (Frontend + Backend)	C	R	R	A	I	A
Reporting & Analytics Module	A	I	A	R	C	R
System Testing	C	R	A	C	I	I
Bug Fixing	R	R	C	I	A	A
Deployment & Hosting Setup	A	I	R	R	C	A
Documentation	A	I	C	A	R	I